

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: II

Solutions for Problem Set 5

(Testing of Hypotheses – Preliminaries, MP test)

Solution 1: Testing $H_0 : \theta = 1$ vs $H_A : \theta = 2$ for $f_\theta(x) = \theta x^{\theta-1}$, $x \in (0, 1)$

Given the critical region:

$$\omega = \left\{ (x_1, x_2) : \frac{1}{2x_1} \leq x_2 \right\}$$

Size of the test (α):

$$\begin{aligned} \alpha &= P_{\theta=1} \left(X_2 \geq \frac{1}{2X_1} \right) \\ &= \int_{1/2}^1 \int_{1/(2x_1)}^1 f_1(x_1) f_1(x_2) dx_2 dx_1 \\ &= \int_{1/2}^1 \int_{1/(2x_1)}^1 (1)(1) dx_2 dx_1 \\ &= \int_{1/2}^1 [x_2]_{1/(2x_1)}^1 dx_1 \\ &= \int_{1/2}^1 \left(1 - \frac{1}{2x_1} \right) dx_1 \\ &= \left[x_1 - \frac{1}{2} \log(x_1) \right]_{1/2}^1 \\ &= (1 - 0) - \left(\frac{1}{2} - \frac{1}{2} \log \left(\frac{1}{2} \right) \right) \\ &= \frac{1}{2} (1 - \log 2) \end{aligned}$$

Power of the test ($1 - \beta$):

$$\begin{aligned} \text{Power} &= P_{\theta=2} \left(X_2 \geq \frac{1}{2X_1} \right) \\ &= \int_{1/2}^1 \int_{1/(2x_1)}^1 f_2(x_1) f_2(x_2) dx_2 dx_1 \\ &= \int_{1/2}^1 \int_{1/(2x_1)}^1 (2x_1)(2x_2) dx_2 dx_1 \\ &= \int_{1/2}^1 2x_1 [x_2^2]_{1/(2x_1)}^1 dx_1 \\ &= \int_{1/2}^1 2x_1 \left(1 - \frac{1}{4x_1^2} \right) dx_1 \\ &= \int_{1/2}^1 \left(2x_1 - \frac{1}{2x_1} \right) dx_1 \\ &= \left[x_1^2 - \frac{1}{2} \log(x_1) \right]_{1/2}^1 \\ &= (1 - 0) - \left(\frac{1}{4} - \frac{1}{2} \log \left(\frac{1}{2} \right) \right) \\ &= \frac{3}{4} - \frac{1}{2} \log 2 \end{aligned}$$

Solution 2: Testing $H_0 : \theta = 2$ vs $H_A : \theta = 1$ for $f_\theta(x) = \frac{1}{\theta}e^{-x/\theta}$, $x > 0$

Simplifying the critical region:

$$\begin{aligned}\frac{f_2(x_1)f_2(x_2)}{f_1(x_1)f_1(x_2)} &\leq \frac{1}{2} \\ \frac{\frac{1}{4}e^{-(x_1+x_2)/2}}{e^{-(x_1+x_2)}} &\leq \frac{1}{2} \\ \frac{1}{4}e^{(x_1+x_2)/2} &\leq \frac{1}{2} \\ e^{(x_1+x_2)/2} &\leq 2 \\ \frac{x_1 + x_2}{2} &\leq \log 2 \\ x_1 + x_2 &\leq \log 4\end{aligned}$$

Let $Y = X_1 + X_2$. Since $X_i \sim \text{Exp}(1/\theta)$, $Y \sim \text{Gamma}(2, \theta)$.

Size of the test (α):

$$\begin{aligned}\alpha &= P_{\theta=2}(Y \leq \log 4) \\ &= \int_0^{\log 4} \frac{1}{4}ye^{-y/2} dy \quad (\text{Substitute } u = y/2, dy = 2du) \\ &= \int_0^{\log 2} \frac{1}{4}(2u)e^{-u}(2du) \\ &= \int_0^{\log 2} ue^{-u} du \\ &= [-ue^{-u} - e^{-u}]_0^{\log 2} \\ &= 1 - e^{-\log 2}(\log 2 + 1) \\ &= \frac{1}{2}(1 - \log 2)\end{aligned}$$

Power of the test ($1 - \beta$):

$$\begin{aligned}\text{Power} &= P_{\theta=1}(Y \leq \log 4) \\ &= \int_0^{\log 4} ye^{-y} dy \\ &= [-ye^{-y} - e^{-y}]_0^{\log 4} \\ &= 1 - e^{-\log 4}(\log 4 + 1) \\ &= 1 - \frac{1}{4}(2\log 2 + 1) \\ &= \frac{3}{4} - \frac{1}{2}\log 2\end{aligned}$$

Solution 3: Testing $H_0 : \theta \leq 30$ vs $H_1 : \theta > 30$ for $X_i \sim N(\theta, 25)$

Given the test $\varphi(\bar{x}) = \mathbb{I}_{\{\bar{x} > c\}}$ and $\beta(\theta) = P_\theta(\bar{X} > c) = 1 - \Phi\left(\frac{c-\theta}{5/\sqrt{n}}\right)$.

For $\theta = 30$:

$$\begin{aligned}\beta(30) &= 0.1 \\ 1 - \Phi\left(\frac{c-30}{5/\sqrt{n}}\right) &= 0.1 \\ \Phi\left(\frac{c-30}{5/\sqrt{n}}\right) &= 0.9 \\ \frac{\sqrt{n}(c-30)}{5} &= 1.282\end{aligned}\tag{1}$$

For $\theta = 35$:

$$\begin{aligned}
\beta(35) &= 0.9 \\
1 - \Phi\left(\frac{c - 35}{5/\sqrt{n}}\right) &= 0.9 \\
\Phi\left(\frac{c - 35}{5/\sqrt{n}}\right) &= 0.1 \\
\frac{\sqrt{n}(c - 35)}{5} &= -1.282
\end{aligned} \tag{2}$$

Adding (1) and (2):

$$\begin{aligned}
\frac{\sqrt{n}}{5}(2c - 65) &= 0 \\
2c &= 65 \\
c &= 32.5
\end{aligned}$$

Subtracting (2) from (1):

$$\begin{aligned}
\frac{\sqrt{n}}{5}(5) &= 2.564 \\
\sqrt{n} &= 2.564 \\
n &= (2.564)^2 \approx 6.57 \implies n = 7
\end{aligned}$$

Solution 4: Deriving MP size- α tests via Neyman-Pearson Lemma

The MP test rejects H_0 if the likelihood ratio $\lambda(x) = \frac{L(\theta_1)}{L(\theta_0)} > k$.

(a) **Exponential Population:** $f_\theta(x) = \frac{1}{\theta}e^{-x/\theta}$

$$\begin{aligned}
\lambda(x) &= \frac{\prod_{i=1}^n \frac{1}{\theta_1} e^{-x_i/\theta_1}}{\prod_{i=1}^n \frac{1}{\theta_0} e^{-x_i/\theta_0}} \\
&= \left(\frac{\theta_0}{\theta_1}\right)^n \exp\left\{\left(\frac{1}{\theta_0} - \frac{1}{\theta_1}\right) \sum_{i=1}^n x_i\right\}
\end{aligned}$$

Since $\theta_1 > \theta_0 \implies \left(\frac{1}{\theta_0} - \frac{1}{\theta_1}\right) > 0$.

$$\lambda(x) > k \implies \sum x_i > c$$

Under H_0 , $\sum X_i \sim \text{Gamma}(n, \theta_0) \implies \frac{2\sum X_i}{\theta_0} \sim \chi_{2n}^2$.

$$\varphi(x) = \mathbb{I}\left\{\sum X_i > \frac{\theta_0}{2} \chi_{2n, \alpha}^2\right\}$$

(b) **Poisson Population:** $f_\theta(x) = \frac{e^{-\theta} \theta^x}{x!}$

$$\begin{aligned}
\lambda(x) &= \frac{e^{-n\theta_1} \theta_1^{\sum x_i} / \prod x_i!}{e^{-n\theta_0} \theta_0^{\sum x_i} / \prod x_i!} \\
&= e^{-n(\theta_1 - \theta_0)} \left(\frac{\theta_1}{\theta_0}\right)^{\sum x_i}
\end{aligned}$$

Since $\theta_1 > \theta_0$, this is monotonically increasing in $\sum x_i$.

$$\varphi(x) = \begin{cases} 1 & \text{if } \sum x_i > c \\ \gamma & \text{if } \sum x_i = c \\ 0 & \text{if } \sum x_i < c \end{cases}$$

(c) **Pareto-like Population:** $f_\theta(x) = \frac{\theta}{x^{\theta+1}}$

$$\begin{aligned}\lambda(x) &= \frac{\theta_1^n \prod x_i^{-(\theta_1+1)}}{\theta_0^n \prod x_i^{-(\theta_0+1)}} \\ &= \left(\frac{\theta_1}{\theta_0}\right)^n \left(\prod_{i=1}^n x_i\right)^{-(\theta_1-\theta_0)}\end{aligned}$$

Taking logarithms:

$$\begin{aligned}n \log \left(\frac{\theta_1}{\theta_0}\right) - (\theta_1 - \theta_0) \sum \log x_i &> \log k \\ -(\theta_1 - \theta_0) \sum \log x_i &> k' \\ \sum \log x_i &< c \quad (\text{since } \theta_1 > \theta_0)\end{aligned}$$

Under H_0 , $2\theta_0 \sum \log X_i \sim \chi_{2n}^2$.

$$\varphi(x) = \mathbb{I}_{\left\{\sum \log x_i < \frac{1}{2\theta_0} \chi_{2n,1-\alpha}^2\right\}}$$

(d) **Normal $N(0, \theta^2)$ Population:**

$$\begin{aligned}\lambda(x) &= \frac{(\sqrt{2\pi}\theta_1)^{-n} \exp\left(-\frac{\sum x_i^2}{2\theta_1^2}\right)}{(\sqrt{2\pi}\theta_0)^{-n} \exp\left(-\frac{\sum x_i^2}{2\theta_0^2}\right)} \\ &= \left(\frac{\theta_0}{\theta_1}\right)^n \exp\left\{\sum x_i^2 \left(\frac{1}{2\theta_0^2} - \frac{1}{2\theta_1^2}\right)\right\}\end{aligned}$$

Since $\theta_1 > \theta_0$, the exponent coefficient is positive.

$$\lambda(x) > k \implies \sum x_i^2 > c$$

Under H_0 , $\frac{\sum X_i^2}{\theta_0^2} \sim \chi_n^2$.

$$\varphi(x) = \mathbb{I}_{\{\sum x_i^2 > \theta_0^2 \chi_{n,\alpha}^2\}}$$

(e) **Normal $N(\theta, 1)$ Population:**

$$\begin{aligned}\lambda(x) &= \frac{\exp\left\{-\frac{1}{2} \sum (x_i - \theta_1)^2\right\}}{\exp\left\{-\frac{1}{2} \sum (x_i - \theta_0)^2\right\}} \\ &= \exp\left\{\frac{1}{2} \sum ((x_i - \theta_0)^2 - (x_i - \theta_1)^2)\right\} \\ &= \exp\left\{(\theta_1 - \theta_0) \sum x_i + \frac{n}{2}(\theta_0^2 - \theta_1^2)\right\}\end{aligned}$$

$$\lambda(x) > k \implies \sum x_i > c \implies \bar{x} > c'$$

$$\varphi(x) = \mathbb{I}_{\left\{\bar{x} > \theta_0 + \frac{1}{\sqrt{n}} z_\alpha\right\}}$$

Solution 5: Testing $H_0 : X \sim f_0$ vs $H_A : X \sim f_1$

We reject H_0 if:

$$\begin{aligned}\frac{f_1(x)}{f_0(x)} &> k \\ \frac{1}{f_0(x)} &> k \\ f_0(x) &< c\end{aligned}$$

Because $f_0(x)$ is symmetric around $1/2$, $f_0(x) < c \implies |X - 1/2| > c'$.

$$P_0(|X - 1/2| > c') = \alpha$$

$$2 \cdot P_0(X < x_0) = \alpha \quad (\text{due to symmetry})$$

$$\int_0^{x_0} 4x \, dx = \frac{\alpha}{2}$$

$$[2x^2]_0^{x_0} = \frac{\alpha}{2}$$

$$2x_0^2 = \frac{\alpha}{2}$$

$$x_0 = \frac{\sqrt{\alpha}}{2}$$

$$\varphi(x) = \mathbb{I}_{\{|x - \frac{1}{2}| > \frac{1 - \sqrt{\alpha}}{2}\}}$$

Solution 6: Testing $f_0(x) = \frac{3x^2}{64}$ **vs** $f_1(x) = \frac{3\sqrt{x}}{16}$

The likelihood ratio is:

$$\begin{aligned} \frac{f_1(x_1)f_1(x_2)}{f_0(x_1)f_0(x_2)} &= \frac{(3/16)^2 \sqrt{x_1 x_2}}{(3/64)^2 (x_1 x_2)^2} \\ &= 16(x_1 x_2)^{-3/2} \end{aligned}$$

Reject H_0 when:

$$16(x_1 x_2)^{-3/2} > k$$

$$(x_1 x_2)^{-3/2} > k'$$

$$\log((x_1 x_2)^{-3/2}) > \log(k')$$

$$-3 \log(x_1 x_2) > c$$

$$-2 \log(x_1^3) - 2 \log(x_2^3) > c'$$

Under H_0 , $F_0(x) = \frac{x^3}{64}$. By PIT, $-2 \log(F_0(X_i)) \sim \chi_2^2$.

$$-2 \log\left(\frac{x_1^3}{64}\right) - 2 \log\left(\frac{x_2^3}{64}\right) \sim \chi_4^2$$

$$-2 \log\left(\frac{(x_1 x_2)^3}{64^2}\right) \sim \chi_4^2$$

$$\log\left(\frac{64^2}{(x_1 x_2)^3}\right) \sim \chi_4^2$$

$$\log\left(\frac{2^{24}}{(x_1 x_2)^6}\right) \sim \chi_4^2$$

Test rejects H_0 if $\log\left(\frac{2^{24}}{(X_1 X_2)^6}\right) > \chi_{4,\alpha}^2$.

Solution 7: Testing $H_0 : \theta = \theta_0$ **vs** $H_A : \theta = \theta_1$ ($< \theta_0$) **for** $f_\theta(x) = \theta x^{\theta-1}$

Likelihood ratio:

$$\begin{aligned} \lambda(x) &= \frac{\theta_1^n \prod x_i^{\theta_1-1}}{\theta_0^n \prod x_i^{\theta_0-1}} \\ &= \left(\frac{\theta_1}{\theta_0}\right)^n \left(\prod_{i=1}^n x_i\right)^{\theta_1-\theta_0} \end{aligned}$$

Taking the logarithm:

$$\begin{aligned} n \log \left(\frac{\theta_1}{\theta_0} \right) + (\theta_1 - \theta_0) \sum \log x_i &> \log k \\ (\theta_1 - \theta_0) \sum \log x_i &> k' \\ \sum (-\log x_i) &> c \quad (\text{since } \theta_1 < \theta_0) \end{aligned}$$

Let $Y_i = -\log X_i$. Under H_0 , $Y_i \sim \text{Exp}(1/\theta_0)$.

$$\begin{aligned} \sum_{i=1}^n Y_i &\sim \text{Gamma}(n, 1/\theta_0) \\ 2\theta_0 \sum_{i=1}^n Y_i &\sim \chi_{2n}^2 \\ \varphi(x) &= \mathbb{I}_{\{-2\theta_0 \sum \log x_i > \chi_{2n, \alpha}^2\}} \end{aligned}$$

Solution 8: Discrete MP Test with $\alpha = 0.04$

Compute the likelihood ratio $\lambda(x) = \frac{f_{H_1}(x)}{f_{H_0}(x)}$:

$$\begin{aligned} \lambda(1) &= 0.06/0.01 = 6 \\ \lambda(2) &= 0.05/0.01 = 5 \\ \lambda(3) &= 0.04/0.01 = 4 \\ \lambda(4) &= 0.03/0.01 = 3 \\ \lambda(5) &= 0.02/0.01 = 2 \\ \lambda(6) &= 0.01/0.01 = 1 \\ \lambda(7) &= 0.79/0.94 \approx 0.84 \end{aligned}$$

Sorting by $\lambda(x)$ descending, we select points for the critical region ω until $\sum P_{H_0}(x) = 0.04$.

$$\begin{aligned} P_{H_0}(X=1) + P_{H_0}(X=2) + P_{H_0}(X=3) + P_{H_0}(X=4) &= 0.01 + 0.01 + 0.01 + 0.01 \\ &= 0.04 \end{aligned}$$

Thus, $\omega = \{1, 2, 3, 4\}$.

$$\varphi(x) = \mathbb{I}_{\{x \in \{1, 2, 3, 4\}\}}$$

Probability of Type II Error (β):

$$\begin{aligned} \beta &= P_{H_1}(X \notin \omega) \\ &= P_{H_1}(X \in \{5, 6, 7\}) \\ &= f_{H_1}(5) + f_{H_1}(6) + f_{H_1}(7) \\ &= 0.02 + 0.01 + 0.79 \\ &= 0.82 \end{aligned}$$